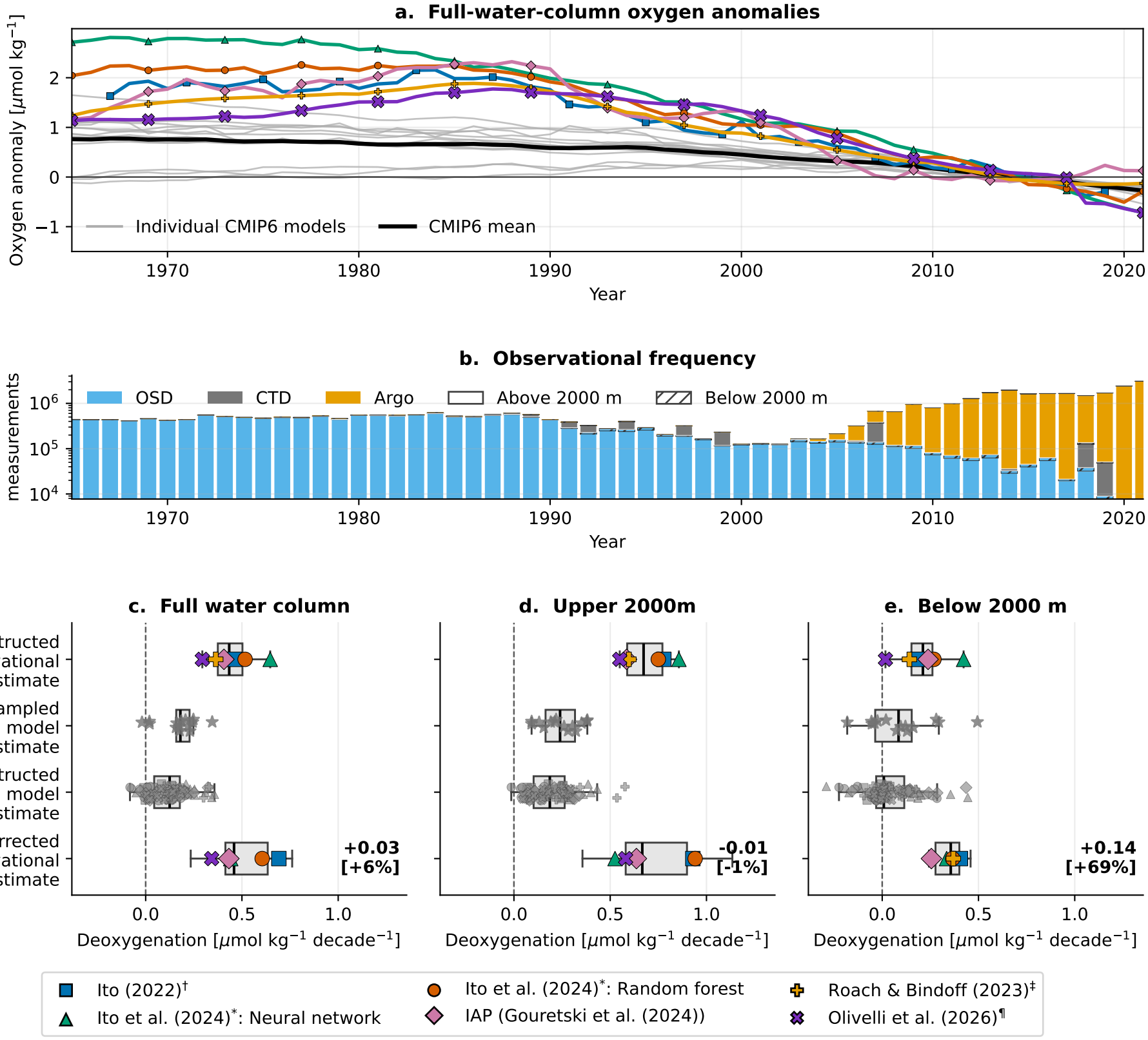

a. Full-water-column oxygen anomalies
Oxygen anomaly [μmol kg⁻¹]
Individual CMIP6 models
CMIP6 mean
Year
b. Observational frequency
Number of measurements
OSD
CTD
Argo
Above 2000 m
Below 2000 m
Year
c. Full water column
d. Upper 2000m
e. Below 2000 m
Reconstructed observational estimate
Fully sampled model estimate
Reconstructed model estimate
Model-corrected observational estimate
+0.03 [+6%]
-0.01 [-1%]
+0.14 [+69%]
Deoxygenation [μmol kg⁻¹ decade⁻¹]
Ito (2022)†
Ito et al. (2024)*: Neural network
Ito et al. (2024)*: Random forest
IAP (Gouretski et al. (2024))
Roach & Bindoff (2023)‡
Olivelli et al. (2026)¶